

Zackary Masri

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SUMMARY

AI/ML Engineer and Data Engineer building agentic AI systems, ML pipelines, and scalable data platforms across Snowflake, Databricks, GCP, and Azure, with deep background in NLP and predictive modeling. Graduate student in applied math and AI/ML. Currently conducting research on Agentic AI and AI-generated code at the Georgia Institute of Technology, with an IEEE publication.

SKILLS

Languages: Python, SQL, Java, C/C++, R, MATLAB, Spark/PySpark

AI / ML: PyTorch, scikit-learn, XGBoost, SHAP, BERT, LangChain, Vertex AI, Gemini, Snowflake Cortex, multi-agent orchestration, RAG

Cloud & Data: GCP, AWS, Azure, Snowflake, Databricks, Docker, Kubernetes, Git, Fabric

BI & App: ThoughtSpot, MicroStrategy, Power BI, Tableau, Looker, Streamlit, React, Node.js, C#/.NET, HTML/CSS

WORK EXPERIENCE

D&I Integrators

Toronto, ON

Lead AI / ML Engineer

Feb 2026 – Present

- Architected a multi-agent AI platform for CVS Health's supply-chain diagnostics on GCP (Vertex AI / Gemini), orchestrating 5 specialized agents (predictive diagnostics, summarization, root-cause analysis, NLP-to-SQL, reasoning) into a conversational analytics layer for planners across 9K+ retail stores, built the supporting Snowflake/BigQuery/Databricks pipeline feeding XGBoost + SHAP models into the agent layer.
- Built and deployed demand-forecasting and production line optimization ML models for Rona, powering production and inventory planning. Owned feature engineering, training, evaluation, and deployment pipelines end-to-end.

Bell Canada

Toronto, ON

Software Developer & Machine Learning Engineer, Speech-to-Text Analytics

Apr 2025 – Feb 2026

- Processed 2M+ call transcripts/month through NLP pipelines (BERT-based sentiment/call-reason classification, entity extraction) on Kubernetes/GCP with Python, SQL, and SAS, validated outputs across 15+ categories using consistency rules and frequency-distribution checks, cutting downstream reporting errors by ~20% and powering MicroStrategy CX dashboards for 500+ stakeholders.
- Generated 10,000+ synthetic call-center transcripts with Google Gemini Flash to expand model training coverage of 30+ rare issue types, boosting training-data diversity by ~35% for agent-assist use cases (next-best-action, risk flags).

Data Science Manager, Digital Insights & Performance

Sep 2023 – Apr 2025

- Deployed 5+ predictive AI/ML models on GCP (Python, SQL/BigQuery, SAS) for Bell's Intelligent Experience Engine, routing 1M+ monthly users into self-serve flows across billing, technical, and account support.
- Engineered 20+ behavioral features across 3 customer segments (clustering + rules) and tuned classification thresholds, saving 100K+ agent hours/year (~7% OPEX reduction, ~\$4M annualized).

AI & Data Engineering Intern

May 2023 – Aug 2023

- Built a Crave TV collaborative-filtering recommendation prototype on GCP/BigQuery with Python (pandas, scikit-learn), evaluated across 500K+ user-item pairs (+15% relevance vs. baseline), decommissioned 3 legacy SQL databases, cutting pipeline latency by ~25% and storage costs by ~\$30K/year.

Avesdo Real Estate Software

Vancouver, BC

Business Intelligence & Full Stack Software Developer Intern

May 2022 – Aug 2022

- Built 8+ web scrapers for real-estate developer sites, normalized 50K+ JSON/XML/HTML records into AWS S3/RDS and delivered QuickSight dashboards tracking 12 launch, inventory, and pricing KPIs, uncovering \$450K+ in pipeline gaps and contributing to ~10% client growth across ON & BC.

Celestica Inc.

Toronto, ON

Data Engineering & Data Analyst Intern

May 2020 – Aug 2021

- Built Azure ETL pipelines and Power BI dashboards tracking 25+ supply-chain KPIs, used SQL, Python/PySpark, and Databricks to process 6.5M+ SAP records, identifying 3 critical bottlenecks with Celonis process mining and improving on-time delivery visibility by ~18%.

PROJECTS

Q-Learning Stock Trading Agent: Built a Q-Learner (+ Dyna-Q) with tuned indicators, buy/sell thresholds, and exploration/decay parameters; evaluated via repeated backtests.

Global Liquidity Crypto Model: Built an ML-driven global-liquidity index and tested BTC/ETH allocation signals using macro features.

EDUCATION

Johns Hopkins University — M.S. Applied & Computational Mathematics (Thesis)

Aug 2025 – Present

Georgia Institute of Technology — M.S. Computer Science (Machine Learning & AI)

Aug 2025 – Present

McMaster University — B.Sc. Mathematics & Statistics; B.Com. Finance & Accounting

Sep 2017 – Apr 2023

Certifications: Snowflake SnowPro Core (COF-C02) • DeepLearning.AI Specializations (ML, DL, NLP, RL, TensorFlow)